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| **Fiscal Year:** | FY 2009 | **Task Last Updated:** | FY 04/07/2009 |
| **PI Name:** | Fike, John R. Ph.D. | | |
| **Project Title:** | Inflammation in the brain after particulate irradiation predisposes the hippocampus to a heightened vulnerability after a secondary insult | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | HUMAN RESEARCH | | |
| **Program/Discipline--Element/Subdiscipline:** | HUMAN RESEARCH--Radiation health | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | (1) **SR**:Space Radiation | | |
| **Human Research Program Risks:** | (1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 94110 | **Congressional District:** | 8 |
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| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2004 Radiation Biology NNH04ZUU005N |
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| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | 1 |
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| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA JSC |
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| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: Received NCE to 5/31/2010 per J. Dardano/JSC (3/2009)<br>NOTE: End date changed to 5/31/2009 per S. Krenek/JSC (8/07)<br>End date changed to 7/14/2009 per JSC info update (10/06) | | |
| **Key Personnel Changes/Previous PI:** | Add: Marta Andres-Mach, Ph.D. Post-doctoral Fellow | | |
| **COI Name (Institution):** | Obenaus, Andre ( Loma Linda University )<br>Noble, Linda ( University of California at San Francisco )<br>Gupta, Nalin ( University of California San Francisco )<br>Rosi, Susanna Ph.D. ( University of California San Francisco ) | | |
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| **Performance Goal Text:** | | | |

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| **Task Description:** | A unique feature of the space radiation environment is the presence of high-energy charged particles including protons and fully ionized atomic nuclei such as 56Fe. These particulate radiations may pose a significant hazard to space flight crews not only during the mission but also later, on Earth, when slow-developing adverse effects finally become apparent. In the CNS, exposure to such radiation may not only directly affect structure/function within the brain, but it may also make tissues more sensitive to a secondary insult such as trauma, stoke or degenerative disease. We have shown that photon and particulate irradiations induce a strong and persistent inflammatory response in the brain that is associated with altered hippocampal neurogenesis and impaired cognitive function. Based on these data we hypothesize that such inflammatory changes, while they may only lead to minor morphologic/functional changes by themselves, will predispose the brain, and in particular hippocampal neurogenesis, to a heightened vulnerability after a secondary insult. Secondly, we hypothesize that a pharmacologic reduction of radiation-induced inflammation will ameliorate the combined effects of radiation and secondary injury on hippocampal neurogenesis. While the nature of possible secondary insults is variable, we have identified traumatic brain injury (TBI) as the choice for the proposed studies. We have identified 4 specific aims to test our hypotheses: 1) determine if inflammatory changes induced by a low dose of proton irradiation predisposes the hippocampus to abnormal neurogenesis after secondary TBI; 2) determine if inflammatory changes induced by a low dose of 56Fe irradiation predisposes the hippocampus to abnormal neurogenesis after secondary TBI; 3) using knock-out mice lacking chemokine receptor 2 (CCR2), determine if a muted monocytic inflammatory response can reverse the influence of low dose proton and/or 56Fe irradiation on TBI-induced alterations in neurogenesis; and 4) determine if pharmacologic intervention after proton and 56Fe irradiation will reduce radiation-induced inflammation and lessen the effects of subsequent TBI. Mice will be irradiated with low radiation doses (0–2 Gy) and 30 days later will subjected to mild/moderate TBI. Neurogenesis and inflammation will be detected using cell specific immunohistochemistry and the effects of irradiation will be quantified using confocal microscopy and unbiased stereology. These studies will address a possible mechanism by which low-dose irradiation may sensitize the brain to secondary injury, and they will also involve a pharmacologic approach to minimize such effects. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Neurogenesis, or the generation of new neurons, continues throughout life in the hippocampal formation of all mammals, including humans. Neurogenesis is associated with hippocampal dependent cognitive functions, and reduced neurogenesis is linked to poorer cognitive function. Radiation alone and traumatic brain injury (TBI) alone both reduce neurogenesis and are correlated with hippocampal dependent cognitive impairments. Both of these insults occur in the context of normal life on earth, either via therapeutic use (irradiation) or as a result of accident or injury (TBI). Given that TBI is a relatively common type of injury, especially in young adults, it is conceivable that individuals receiving therapeutic irradiation for treatment of brain tumors, head and neck tumors, etc., could experience combined injury. Additionally, given the state of the world at this time, combined injury in a nuclear battlefield or in an uncontrolled radiation exposure such as a dirty bomb, is a possible outcome of a terrorist event. As a consequence, understanding the mechanisms associated with radiation, TBI alone or combined injury is essential for the development of suitable and effective countermeasures or treatments. |
| **Task Progress:** | As detailed in my last progress report, we experienced some significant technical problems associated with anesthesia that impacted our studies and slowed us down. That and lack of availability of specific mutant mice that were originally proposed, required a change of focus in our studies. In the past funding year, therefore, in addition to our neurogenesis endpoints (as originally proposed), we added new experimental approaches to address Arc expression and the assessment of behavioral function. Over the time period from the last progress report, we finished some analyses (see below) of neurogenesis from animals irradiated in the last funding year. In the past year we completed 2 runs at BNL; both involved 2-month old male wild type C57BL/6 mice. Each treatment group consisted of 10 mice. Mice were anesthetized with isoflurane gas and given brain only doses of 56Fe irradiation (0, 0.5, and 1 Gy). Two months after irradiation mice were anesthetized with isoflurane gas and received focal TBI to the left (ipsilateral) hemisphere; we have previously published data documenting the methodological details. Sham-treated animals (controls) were subjected to the same surgical procedure but were not injured. Two weeks following TBI mice received a single injection of BrdU (100 mg/kg) for 5 consecutive days to label proliferating cells. Three weeks after the last BrdU injection, mice underwent 5 days of training and testing in the Morris water maze. One hour after the last hidden platform session mice were required to complete a test to assess spatial memory retention and to induce Arc. Thirty minutes after the water maze test, mice were anesthetized and decapitated. The brains were frozen immediately and later were sectioned for the histologic assessment of neurogenesis, Arc expression and neuroinflammation (numbers of activated microglia), as previously described by us. The analyses of neurogenesis were done on tissues from animals irradiated during the previous funding period. The analysis of one of the runs performed in the current funding year is nearing completion and the data are described here below. The animals from the second BNL run from the past year are post-TBI and will undergo behavioral training and testing in a few weeks. In both of the recent studies, all animals tolerated the irradiation procedures; none were lost during irradiation or shipping from BNL to UCSF. Based on our experience with TBI, we expected ~ a 5% mortality due to the trauma itself, but we lost only 2-3 animals out of the 80 entered into the studies. BrdU: We assessed the survival of newly born, BrdU+ cells (independent of phenotype), and all positively labeled cells were counted in each hemisphere using a stereologic approach. After 0.5 Gy only, there was a significant decrease in the numbers of BrdU+ cells in the ipsilateral hemisphere relative to sham treated controls. No significant alterations were seen in the contralateral hemisphere. TBI only increased the number of BrdU+ cells relative to mice that received sham treatment, but the change was not significant; no significant changes were seen in the contralateral hemisphere. After combined injury, in the ipsilateral hemisphere the numbers of BrdU+ cells were not different from sham treated mice but were lower than after trauma alone. No significant changes were seen in the contralateral hemisphere. These data suggest that at least from the perspective of newly born cells, previous irradiation impacted the total number of newly born cells but only relative to trauma alone.<br><br>Newly born neurons: In sham treated controls, the numbers of newly born neurons averaged about 500 cells/hemisphere. Relative to controls, irradiation alone significantly reduced new neuron production in the ipsilateral hemisphere but only minimally in the contralateral hemisphere. In the ipsilateral hemisphere there was a non-significant increase in newly born neurons after trauma alone and in the contralateral hemisphere there was a nons-ignificant decrease. After combined injury the numbers of newly born neurons were slightly decreased relative to trauma alone but an increase relative to irradiation alone. In the contralateral hemisphere the number of newly born neurons after combined injury was about the same as that seen in sham treated mice. Based on these data, radiation did not appear to |

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| | lead to a heightened vulnerability to a second insult.<br><br>Newly born astrocytes: The numbers of newly born astrocytes before and after single or combined injury were about the same in both hemispheres. Radiation induced a decrease, trauma alone lead to an increase and combined injury showed more newly born astrocytes than radiation alone but less than trauma alone.<br><br>Activated microglia: No data are presently available, but analysis is underway. Arc protein: In both hemispheres of sham-treated mice, the fractions of dentate neurons expressing Arc protein averaged about 3.5-4.0%. The total number of granule cell neurons averages about 250,000 cells in C57BL6 mice, so modest changes in Arc mRNA and Arc protein can represent a relatively large number of cells in the dentate gyrus and could be extremely important for proper hippocampal function. In our study, after 0.5 Gy alone there was a slight, but insignificant, reduction in the fraction of neurons expressing Arc protein. However, after trauma alone, in both ipsilateral and contralateral hemispheres there were significant reductions in the fractions of neurons expressing Arc protein. These reductions were not due to neuronal cell loss. After combined injury there were no differences relative to sham-treated mice, again suggesting that a previous exposure to irradiation did not make the brain more susceptible to a secondary insult. The Arc protein data even suggest that irradiation may have a protective effect against subsequent traumatic injury. Measures of Arc mRNA are currently ongoing.<br><br>Behavioral performance: There were no effects of radiation, trauma, or combined radiation/trauma on ability of the mice to locate the visible or hidden platform in the water maze. However, treatment effects were seen when spatial memory retention was assessed in the probe trial (no platform) following training to locate the hidden platform. While sham treated mice and mice receiving radiation only showed spatial memory retention and spent most of their time searching in the part of the pool previously containing the hidden platform, mice that received trauma alone did not. This showed that unilateral cortical impact injury induced substantial hippocampus-dependent cognitive impairments. When traumatic brain injury occurred 30 days after irradiation, performance in the water maze was not different from sham-treated animals, suggesting that low dose irradiation had a protective effect in the context of a subsequent traumatic injury. |
| **Bibliography Type:** | Description: (Last Updated: 04/11/2018) |
| **Abstracts for Journals and Proceedings** | Fike JR, Rosi S, Fishman K, Baure J, Andres-Mach M, Obenaus A. "Effects of 56Fe irradiation and traumatic brain injury hippocampal neurogenesis." 19th Annual NASA Space Radiation Investigators' Workshop, Philadelphia, PA, June 30 - July 2, 2008.<br>Proceedings from 19th Annual NASA Space Radiation Investigators' Workshop, June 30 - July 2, 2008. , Jul-2008 |
| **Articles in Peer-reviewed Journals** | Fike JR, Rola R, Limoli CL. "Radiation response of neural precursor cells." Neurosurg Clin N Am. 2007 Jan;18(1):115-27, x. Review. PMID: 17244559 , Jan-2007 |
| **Articles in Peer-reviewed Journals** | Fike JR, Rosi S, Limoli CL. "Neural precursor cells and central nervous system radiation sensitivity." Semin Radiat Oncol. 2009 Apr;19(2):122-32. PubMed PMID: 19249650. , Apr-2009 |